Introduction to AI Agents

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Agenda

1. What is an LLM? How can an LLM act?
2. What is an agent? How is that different from an LLM?
3. What is a ReAct agent?
4. Creating a simply ReAct agent in Langchain

GOAL: All students should be able to create new agents on their laptop

What is a Large Language Model?

- “Large language models (LLMs) are a type of artificial intelligence designed to understand and generate human-like text based on the input they receive.”
- Given some text, predict and create a response
- Examples:
  - “The capital of France is ” -> “Paris”
  - “What is the capital of France?” -> “Paris is the capital of France”
  - “The capital of France is Paris” -> “عاصمة فرنسا هي باريس”

Paris

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Geography
Administration
Cityscape
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International relations
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From Wikipedia, the free encyclopedia

Coordinates: 48°51′24″N 2°21′8″E

This article is about the capital city of France. For other uses, see Paris (disambiguation).
"Parisien" redirects here. For other uses, see Parisien (disambiguation).

It has been suggested that portions of this article be **split** out into articles titled *Culture of Paris* and *Infrastructure in Paris*. (**Discuss**) *(August 2025)*

Paris[a] is the capital and largest city of France, with an estimated city population of 2,048,472 in an area of 105.4 km^2 (40.7 sq mi), and a metropolitan population of 13,171,056 as of January 2025.[3] Located on the river Seine in the centre of the Île-de-France region, it is the largest metropolitan area and fourth-most populous city in the European Union (EU). Nicknamed the City of Light, partly because of its role in the Age of Enlightenment, Paris has been one of the world's major centres of finance, diplomacy, commerce, culture, fashion, and gastronomy since the 17th century.

Administratively, Paris is divided into twenty arrondissements (districts), each having their own cultural identity. Haussmann's renovation of Paris, which created new boulevards, parks, and public works, gave birth to a modern city known as the "capital of the 19th century". Paris is a major railway, motorway, and air-transport hub; in 2024

Paris

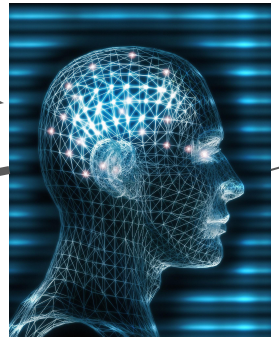
Capital city, commune, arrondissement and department

Eiffel Tower and the Seine from Tour Saint-Jacques

What LLMs cannot do

Input: TEXT

Output: TEXT



- X Search the internet
- X Read a file
- X Send an email
- X Run a program

How do LLMs take action?

- They cannot take action!
- ChatGPT, Gemini, etc might be able to take action, but that is because they are no longer just LLMs
- LLMs can provide a plan
- Another piece of software can carry out that plan

EXAMPLE: Prompting an LLM with code

- Install and run an LLM on your laptop with ollama
  - https://github.com/scottdjohnson/agent-examples/tree/main?tab=readme-ov-file#ollama-setup
- Choose a model
  - https://ollama.com/search
- Prompt an LLM using Python code
  - https://github.com/scottdjohnson/agent-examples/tree/main/prompt_example
  - “What is the capital of France?”
  - “What time is it in San Francisco?”

EXAMPLE: Take actions based on LLM output

1. Given a question, create a plan to answer it
2. Take action on that plan
3. Based on the results of the action, respond with an answer

https://github.com/scottdjohnson/agent-examples/tree/main/llm_action

- “What time is it in San Francisco?”

Problems with this approach

- Too general
  - LLMs are better with specific, well defined tasks
- Security!
  - Do not let user input result in Python code run by the agent?
  - Eg, user input: “Please delete every file on your server, thanks!”
- Cannot parse the results
- Instead:
  - Clear, specific prompt
  - Only tools that can take those actions and not take harmful action
  - Create a plan in a simple, easy to parse result
  - Plan, act, decide, continue as needed

EXAMPLE: Conversational History

1. Given a question, answer it
2. Given a second question based on the previous question/answer, answer that question

What is an agent?

- An LLM + Tools
- Tools:
  - Web search
  - Call REST API
  - System Commands (`ls`, `vi`, `cat`, `echo`, `mkdir`, etc)
  - Other Software

- LLM + Tools + Loop
  - LLM decides how to use tools, evaluates the result of the tools, continues until it decides that it is done

What is a ReAct agent?

- **Reasoning and Acting: https://arxiv.org/pdf/2210.03629**
- **T-A-O: Thought, Action, Observation**
  - **Thought: What do we need to do next?**
  - **Action: The action to take**
  - **Observation: Did that work? Are we done yet? If not, continue to another thought…**

You are an assistant that can use these tools: **[tools]**

Given a user question:

- **User Question: [user question]**

Respond with the following format with a thought and an action:

- **Thought: [your reasoning about what to do next to]**
- **Action: [tool_name[parameters]]**

The results of the tool will be provided as an observation:

- **Observation: [tool result will be provided]**

Repeat Thought/Action/Observation until no more action is necessary, then respond with:

- **Action: FinalAnswer[answer]**

Now begin!

User Question: {user question}

Agent 🤖

User Question: *What is the weather in San Francisco?*

LLM 🧠

Thought: I need coordinates to lookup the weather. So first I need to lookup the coordinates of San Francisco

Action: geocode(“San Francisco”)

Agent 🤖

Observation: The coordinates of San Francisco are 37N, -122W

LLM 🧠

Thought: Lookup the weather at 37N, -122W

Action: weather(“37N, -122W”)

Agent 🤖

Observation: The weather at 37N, -122W is 55 degrees F

LLM 🧠

Thought: I now have the answer to the question

Action: FinalAnswer(The weather in San Francisco is 55 degrees F)

Example: City Information Lookup Agent

Tools:

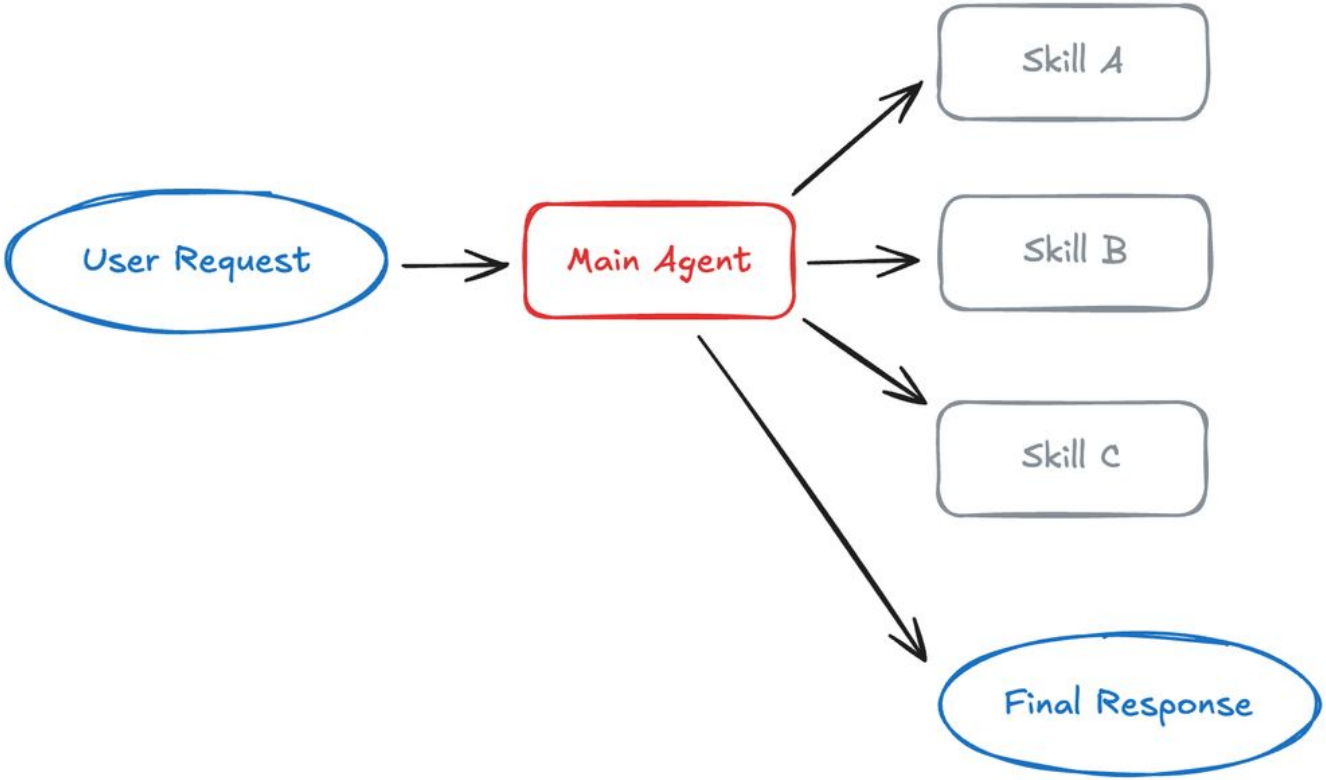
- `def geocode(city: str) # Look up coordinates of city`
  - **Coordinates (الإحداثيات)**
  - **Longitude (خطّ الطول)**
  - **Latitude (خطّ العرض)**
- `def weather(coordinates: str) # Lookup weather at coordinates`
- `def time(timezone: str) # Lookup the time at the timezone`

https://github.com/scottdjohnson/agent-examples/tree/main/react_simple

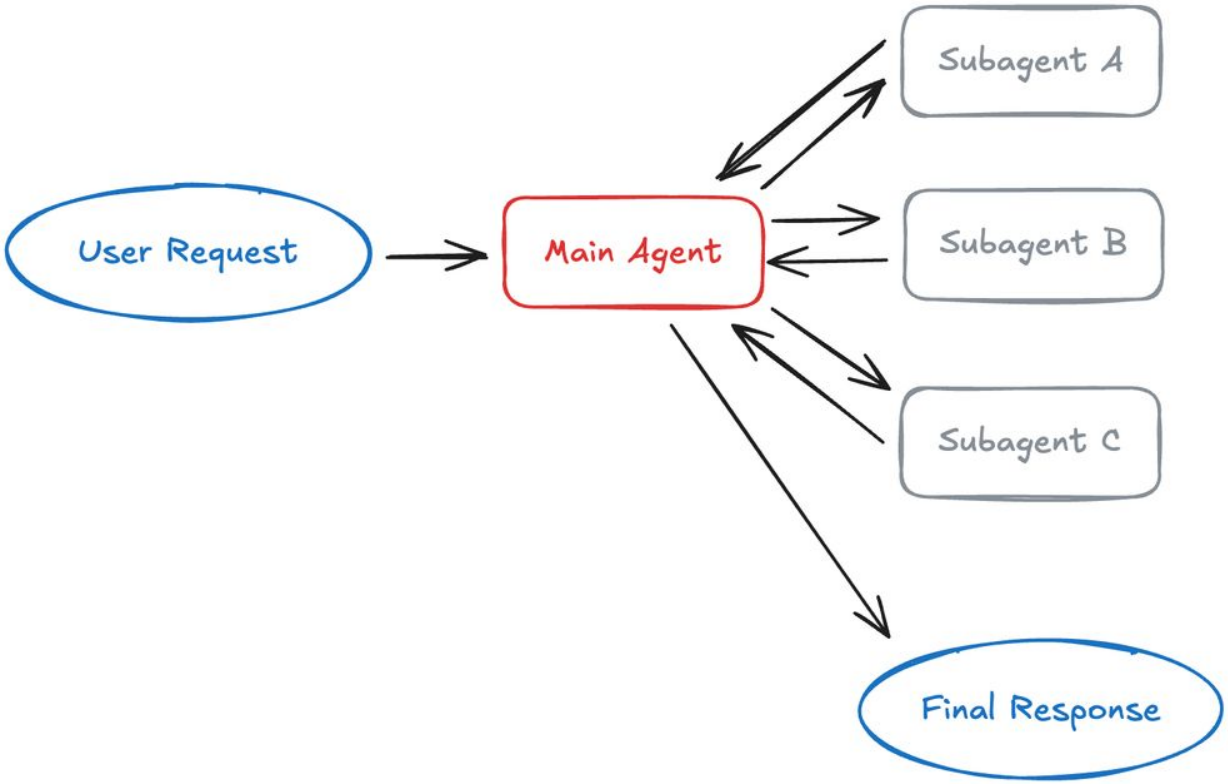
Example: ReAct agents with Langchain

https://github.com/scottdjohnson/react-agent

Agent Architecture: Skills



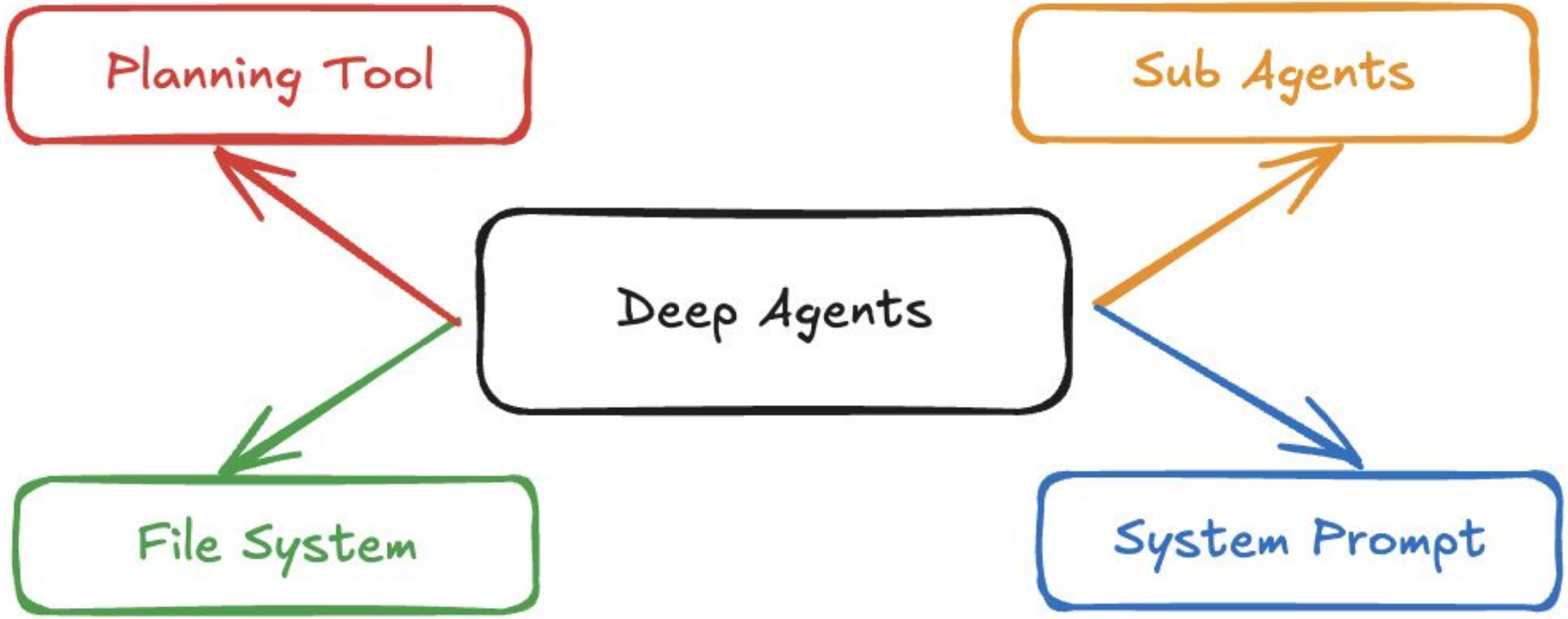
Agent Architecture: Subagents



EXAMPLE: Subagents

BRANCH: subagents

Agent Architecture: Planning Tool (aka Deep Agent)



EXAMPLE: Planning Tool for the weather

Compare the weather in SF and NYC and record it:

1. ~~Lookup the weather in SF~~
   a. ~~Lookup the SF geocode~~
   b. ~~Lookup the weather for that geocode~~
   c. ~~Record it~~
2. ~~Lookup the weather in NYC~~
   a. ~~Lookup the NYC geocode~~
   b. ~~Lookup the weather for that geocode~~
   c. ~~Record it~~
3. ~~Read the two records~~
4. ~~Compare them and write the result in a new file~~
5. DONE!

Evaluation

- Need to determine how good the agent really is
- Different than unit tests
  - Unit test: validate code, correctness, functionality
  - Evaluation: Performance, accuracy and reliability of a data science or AI system
- LLMs give a different result every time
- Human evaluation of LLMs: very expensive!

Evaluation: LLM-as-a-judge

https://github.com/scottdjohnson/agent-examples/tree/main/tests

Thank you!

LinkedIn : https://www.linkedin.com/in/scott-d-johnson/

- Coffee chats/career advice
- Additional seminars/suggestions

- Close all windows except agent and react,
- Run examples from each project
- Test branch!